

Curriculum Vitae: Thomas Vogel

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Education

2018 Ph.D. in Computer Science, Hasso Plattner Institute/University of Potsdam (summa cum laude)

2008 Diploma in Information Systems, University of Bamberg (graduated with distinction)

Positions

2022-present Postdoctoral Researcher, Humboldt-Universität zu Berlin

2021-2022 Stand-in Professor for Empirical Software Engineering (W3), Paderborn University

2018-2021 Postdoctoral Researcher, Humboldt-Universität zu Berlin

2017-2018 Research Assistant, Humboldt-Universität zu Berlin

2013-2017 Research Assistant, Hasso Plattner Institute/University of Potsdam

2008-2013 Ph.D. Student (Scholarship), Hasso Plattner Institute/University of Potsdam

Selected Awards

2025 **Distinguished Lecturer** at the Computer Science Department of the Humboldt-Universität zu Berlin for outstanding teaching performance with the course “Software Engineering II” based on student questionnaires

2025 **SEAMS 10-Year Most Influential Paper Award** for “[Software Engineering meets Control Theory](#)” (SEAMS 2025; with Antonio Filieri, Martina Maggio, Konstantinos Angelopoulos, Nicolas D’Ippolito, Ilias Gerostathopoulos, Andreas Hempel, Henry Hoffmann, Pooyan Jamshidi, Evangelia Kalyvianaki, Cristian Klein, Filip Krikava, Sasa Misailovic, Alessandro Papadopoulos, Suprio Ray, Amir Sharifloo, Stepan Shevtsov, Mateusz Ujma)

2024 **SEAMS 2024 Best Paper Award** for “[Formal Synthesis of Uncertainty Reduction Controllers](#)” (with Marc Carwehl, Calum Imrie, Genáina Rodrigues, Radu Calinescu, and Lars Grunske)

2021 **Top 1% of SoSyM Reviewers** of the Journal of Software and Systems Modeling (SoSyM)

2018 **Facebook Testing and Verification Research Award** (acceptance rate <7%)

2017 **Karsten Schwan Best Paper Award** for “[Efficient Utility-Driven Self-Healing Employing Adaptation Rules for Large Dynamic Architectures](#)” (ICAC 2017; with Sona Ghahremani and Holger Giese)

2012 **SEAMS 2012 Best Paper Award** for “[A language for feedback loops in self-adaptive systems: Executable runtime megamodels](#)” (with Holger Giese)

Grants

2025 Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) grant for “Controlling Search-Based Test Case Generation and Program Repair”

2018 Facebook Testing and Verification Research Award for “Self-Adaptive Search for Sapienz (S3)”

Mentoring Ph.D. Students

Current Hannes Ueck and Marc Carwehl

2026 Dr. Sebastian Müller (“Metamorphic Testing for Scientific Software Using Geometrically Representable Input Data”)

2023 Dr. Arut Prakash Kaleeswaran (“Explanation of the Model Checker Verification Result”)

2023 Dr. Sona Ghahremani (“Incremental Self-Adaptation of Dynamic Architectures Attaining Optimality and Scalability”, summa cum laude)

Teaching (selected courses)

Lecturer “AI Controlled Software Testing (2025) • “Software Engineering II” (2017–2025, with Lars Grunske) • “Adaptive Systeme” (2023, with Marc Carwehl) • “Software Quality Assurance” (2022) • “Search-Based Software Engineering” (2022) • “Learning, Optimization, and Assurances for Self-Adaptive Systems” (2022) • “Software Engineering for Self-Adaptive Systems” (2021) • “Requirements Engineering and Software Architecture” (2021)

Assistant “Software Engineering” (2019–2025) • “Methoden und Modelle des Systementwurfs” (2022–2025) • “Modellbasierte Entwicklung eingebetteter Systeme” (2021) • “Software Verification” (2020)

Steering Committee Memberships

2023–present ACM/IEEE International Conference on Software Engineering for Adaptive and Self-Managing Systems (SEAMS) (nominated)

2020-2022 International Symposium on Search-Based Software Engineering (SSBSE) (elected)

Editorial Boards

Since 05/2025 Associate Editor of ACM Transactions on Autonomous and Adaptive Systems (TAAS)

Since 01/2014 Editor-in-Chief of self-adaptive.org, the central website of the research community on software engineering for self-adaptive systems

Organization of Recent Scientific Meetings

2026	Review Process Co-Chair	ACM/IEEE International Conference on Automated Software Engineering (ASE)
2025	Program Co-Chair	Research Track, ACM/IEEE International Conference on Software Engineering for Adaptive and Self-Managing Systems (SEAMS)
2023	Workshop Co-Chair	Gesellschaft für Informatik (GI) conference on Software Engineering (SE)
2023	Program Co-Chair	HOP Track, International Symposium on Search-Based Software Engineering (SSBSE)
2022	Program Co-Chair	Challenge Track, International Symposium on Search-Based Software Engineering (SSBSE)
2021	Program Co-Chair	DEI Track, 5th European Conference on Software Architecture (ECSA)
2021	Program Chair	Artifact Track, ACM/IEEE International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS)

Co-Organizer of three GI Dagstuhl Seminars: Explainable Software for Cyber-Physical Systems (2019), Software Engineering for Intelligent and Autonomous Systems (2018), Software Engineering for Self-Adaptive Systems (2014).

Selected Program Committee Memberships

ACM/IEEE Int'l Conference on Software Engineering (ICSE): Research Track (2024, 2027), NIER Track (2020)

ACM/IEEE Int'l Conference on Automated Software Engineering (ASE): Research Track (2025), NIER Track (2022)

IEEE Int'l Conference on Software Testing, Verification and Validation (ICST): Research Track (2022–2025), Short Papers, Vision and Emerging Results Track (2026), Testing Tools and Data Showcase Track (2026)

ACM/IEEE Int'l Conference on Software Engineering for Adaptive and Self-Managing Systems (SEAMS): Research Track (2018–2024, 2026–2027), Artifacts (2015, 2023)

ACM/IEEE Int'l Conference on Automation of Software Test (AST): 2025–2027

ACM Genetic and Evolutionary Computation Conference (GECCO): 2019–2023

IEEE Int'l Conference on Autonomic Computing and Self-Organizing Systems (ACSOS): Research Track (2021–2022, 2026), Research Track Senior PC (2023), Artifacts (2026), Posters & Demos (2020–2021)

Symposium on Search-Based Software Engineering (SSBSE): Research Track (2021–2023, 2025), NIER Track (2020, 2021)

Five Representative Publications

- [1] Marc Carwehl, Calum Imrie, **Thomas Vogel**, Genáina Rodrigues, Radu Calinescu, and Lars Grunske. “Formal Synthesis of Uncertainty Reduction Controllers”. In: *Proceedings of the 19th International Conference on Software Engineering for Adaptive and Self-Managing Systems*. SEAMS '24. **Best Paper Award**. ACM, 2024, pp. 2–13. DOI: [10.1145/3643915.3644095](https://doi.org/10.1145/3643915.3644095).
- [2] João Paulo Costa de Araujo, Genáina Nunes Rodrigues, Marc Carwehl, **Thomas Vogel**, Lars Grunske, Ricardo Caldas, and Patrizio Pelliccione. “Explainability for Property Violations in Cyberphysical Systems: An Immune-Inspired Approach”. In: *IEEE Software* 41.5 (2024), pp. 43–51. DOI: [10.1109/MS.2024.3387289](https://doi.org/10.1109/MS.2024.3387289).
- [3] Arut Prakash Kaleeswaran, Arne Nordmann, **Thomas Vogel**, and Lars Grunske. “A User Study for Evaluation of Formal Verification Results and their Explanation at Bosch”. In: *Empirical Softw. Engineering (EMSE)* 28.125 (2023). DOI: [10.1007/s10664-023-10353-4](https://doi.org/10.1007/s10664-023-10353-4).
- [4] **Thomas Vogel**, Chinh Tran, and Lars Grunske. “A Comprehensive Empirical Evaluation of Generating Test Suites for Mobile Applications with Diversity”. In: *Information and Software Technology* 130 (2021), p. 106436. DOI: [10.1016/j.infsof.2020.106436](https://doi.org/10.1016/j.infsof.2020.106436).
- [5] **Thomas Vogel** and Holger Giese. “Model-Driven Engineering of Self-Adaptive Software with EUREMA”. In: *ACM Transactions on Autonomous and Adaptive Systems* 8.4 (2014), 18:1–18:33. DOI: [10.1145/2555612](https://doi.org/10.1145/2555612).

All publications: [Homepage](#) • [DBLP](#) • [Google Scholar](#) • [ORCID: 0000-0002-7127-352X](#)